Cold Diffusion

do diffusion models actually need noise?

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RE-IMPLEMENTATION

Bansal et al. · arXiv:2208.09392

HOT ●
Gaussian noise
standard DDPM

↓ THE PAPER'S QUESTION ↓
does the noise matter?

● COLD
Gaussian blur
our scope ✦

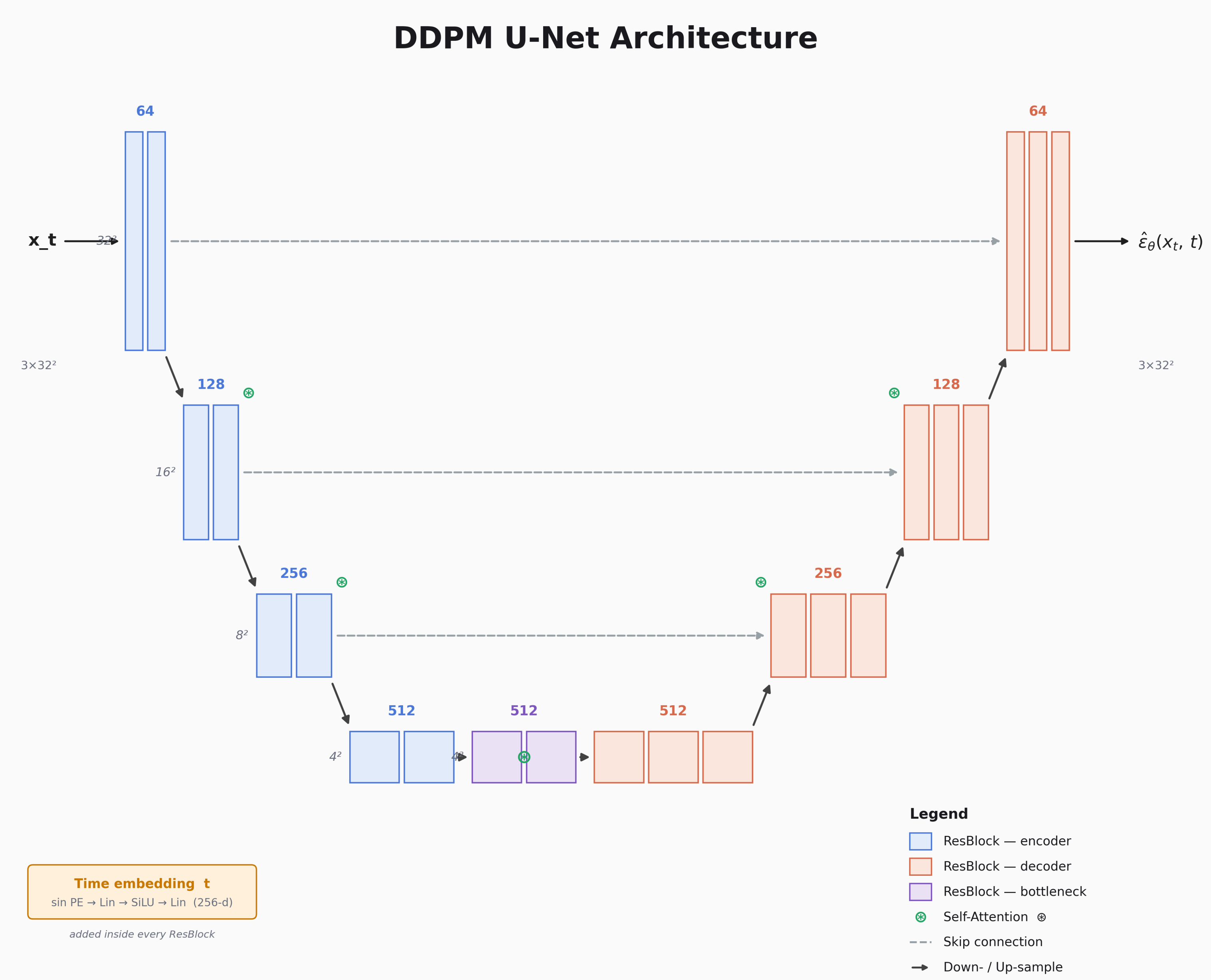
1. Motivation

→ Diffusion ≡ Gaussian noise — by construction.
→ What if we strip the noise out entirely?
→ Replace it with a deterministic blur.
→ Does it still generate? **The paper says yes.**

2. Method

U-NET + ℓ₁

$R_\theta(D(x,t), t) \approx x \quad \cdot \quad D = \text{blur}$



→ U-Net: ResBlocks + attention + skip
→ Sampling: Algorithm 2 (improved)
→ Training: Adam, lr 2e-5

3. Datasets

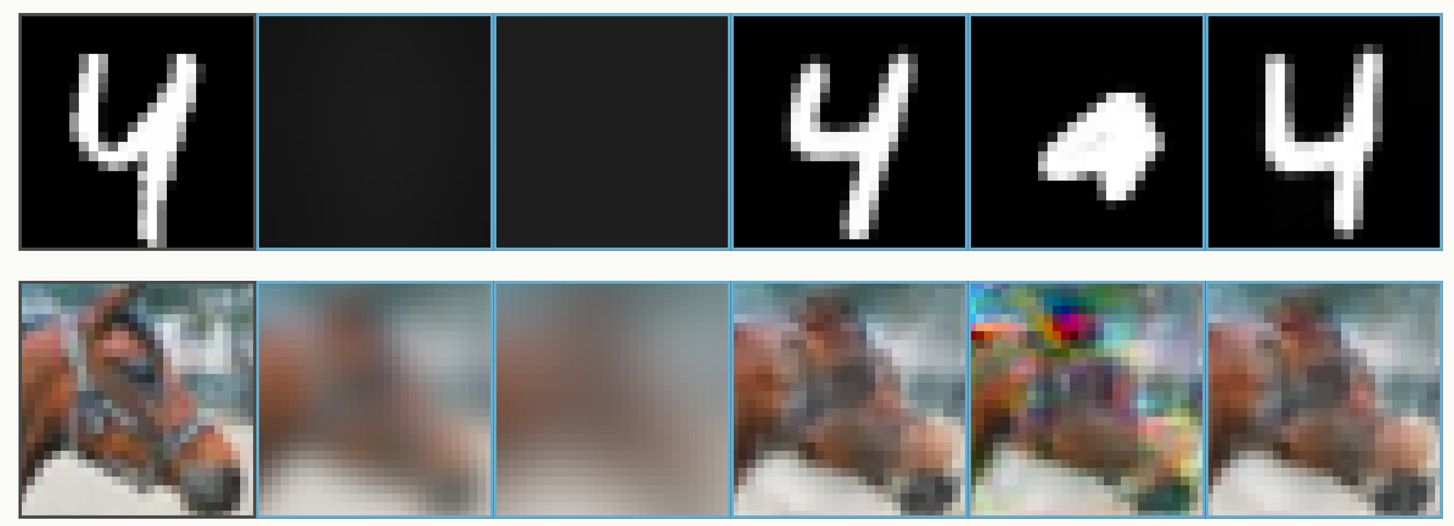
MNIST	CIFAR-10
28×28 gray	32×32 RGB

★ Forward freeze → reverse thaw

FORWARD · DEGRADE x_0 | REVERSE · RESTORE FROM x_T

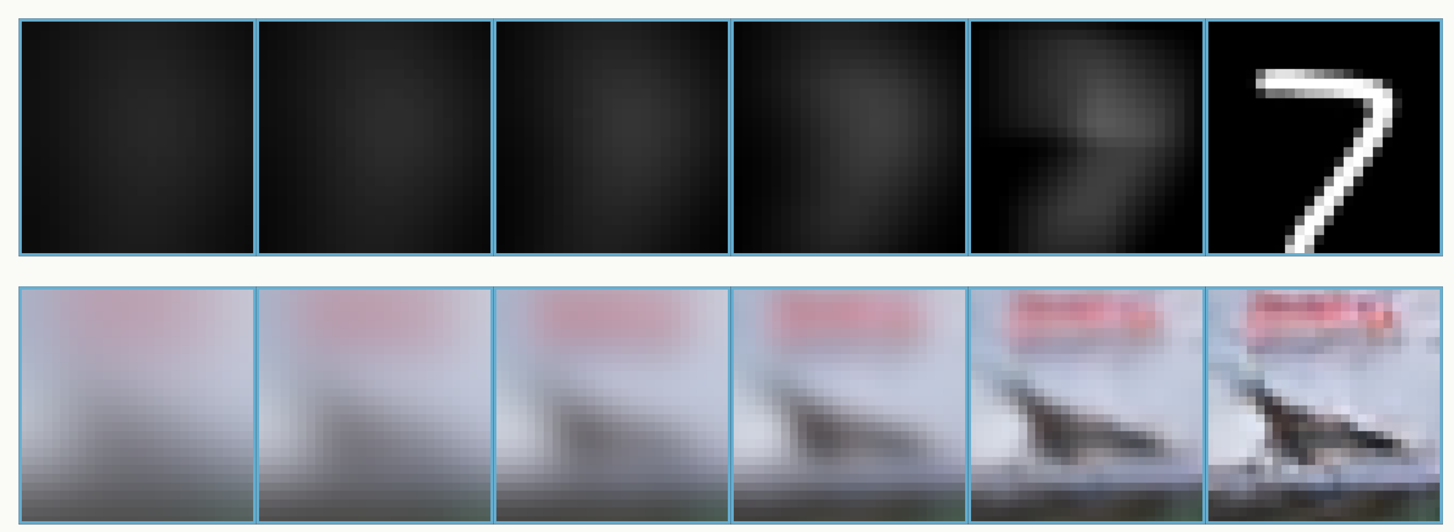
Original x_0 · Mid-blur $D(x_0, T/2)$ · Fully blurred $D(x_0, T) = x_T$ · Direct $R(x_T, T)$ · Algo 1 naive sampler · Algo 2 improved sampler

MNIST
CIFAR-10



SAMPLING TRAJECTORY · ALGO 2

MNIST · Algo 2
CIFAR-10 · Algo 2



Top: forward $x_0 \to x\_T$; reverse compares Direct · Algo 1 · Algo 2. Bottom: Algo 2 running estimate (s = 5→0).

✦ Quantitative results

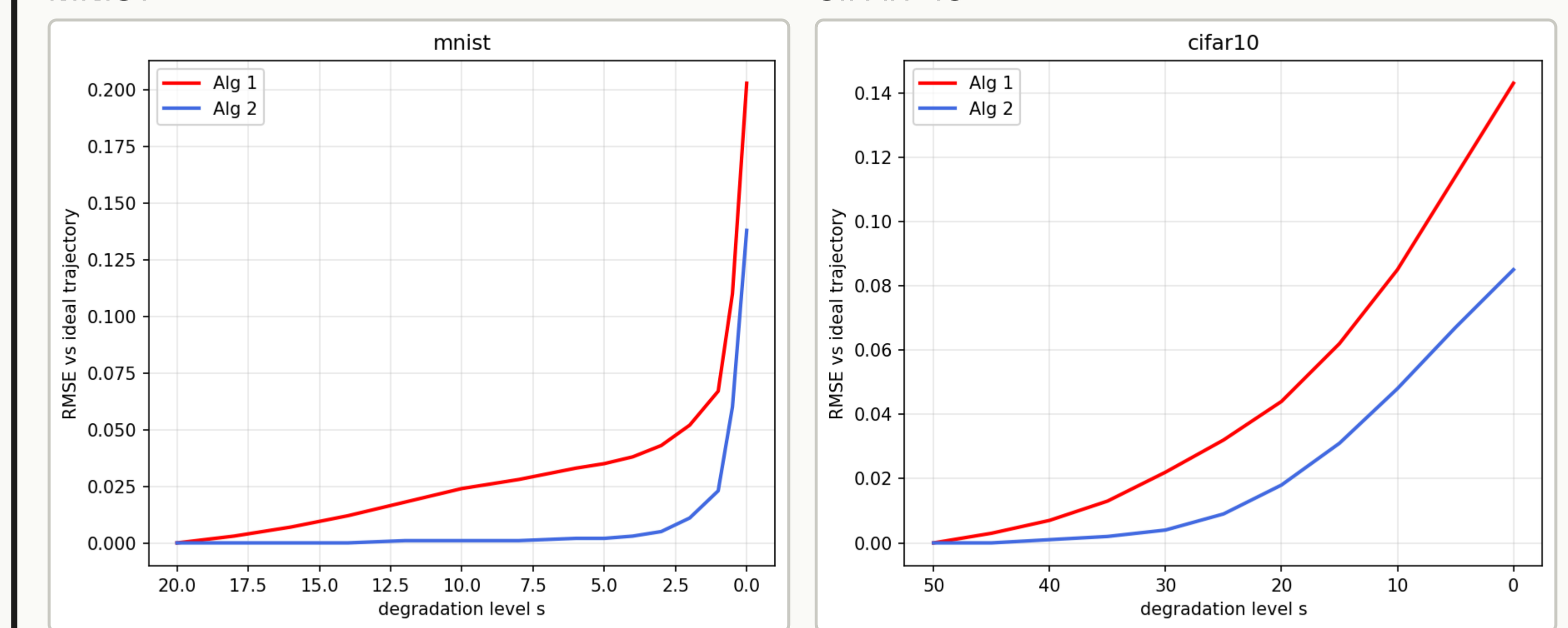
FID / SSIM / RMSE · HELD-OUT TEST SET · LOWER IS BETTER

Dataset	Method	Ours			Paper		
		FID	SSIM	RMSE	FID	SSIM	RMSE
MNIST	Degraded (x_T)	398.14	0.0141	0.2866	368.56	0.1780	0.2310
	Direct R(x_T,T)	27.04	0.8458	0.1358	4.05	0.8230	0.1140
	Alg 1 naive iter.	41.16	0.6234	0.2374	—	—	—
	Alg 2 improved	22.69	0.7537	0.1780	4.33	0.8200	0.1150
CIFAR-10	Degraded (x_T)	313.65	0.3562	0.1359	358.99	0.2790	0.1460
	Direct R(x_T,T)	58.33	0.6736	0.0810	169.94	0.4200	0.1520
	Alg 1 naive iter.	96.42	0.5188	0.1205	—	—	—
	Alg 2 improved	55.75	0.6634	0.0826	152.76	0.4110	0.1550

4. Results

DRIFT OVER SAMPLING

MNIST · CIFAR-10



RMSE vs ideal trajectory. Alg 1 (red) drifts; Alg 2 (blue) stays close to the manifold.

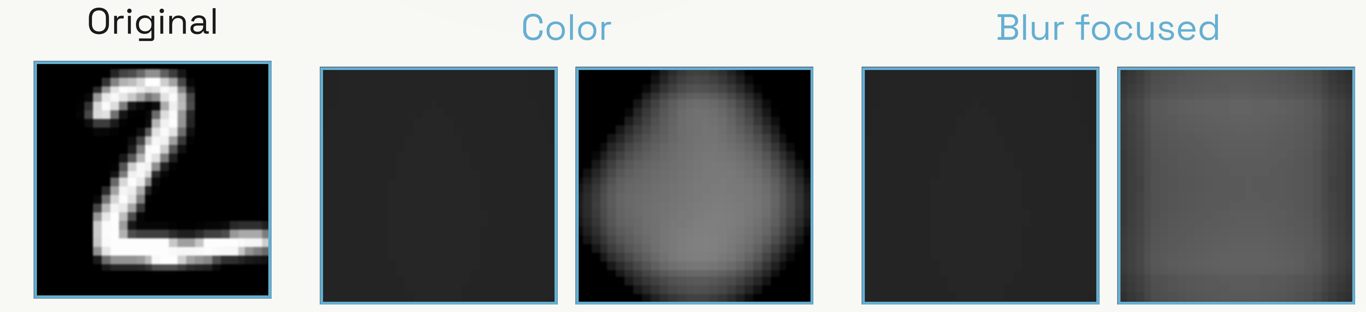
QUANTITATIVE DRIFT / STABILITY

→ Per-step drift confirms the compounding error theory of §3.3.
→ Alg 1 accumulates restoration error; Alg 2's self-correcting update keeps drift ≈ 0.
→ Alg 2 < Alg 1 at every degradation level on both datasets.

5. Further experiments / exploration

→ Random color removed at each timestep
→ Focused gaussian blur around brighter points

Original · Color · Blur focused



Method	Color			Blur focused		
	FID	SSIM	RMSE	FID	SSIM	RMSE
Degraded (x_T)	398.14	0.0141	0.2866	398.14	0.0141	0.2866
Direct R(x_T,T)	314.47	0.0276	0.3457	385.89	−0.0094	0.4333
Alg 1 naive iter.	358.55	0.0291	0.5448	345.62	0.0389	0.3101
Alg 2 improved	302.72	0.2732	0.2530	347.28	0.0155	0.2849